

Atul Pokharel

AI Engineer | Forward-Deployed AI Engineer | AI Implementation Engineer

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Professional Summary

AI Engineer with 7+ years of experience building and implementing LLM applications, RAG systems, AI agents, workflow automation tools, backend services, and production AI prototypes. Strong at working across both technical and business environments: understanding user workflows, designing AI solutions, building Python/FastAPI services, integrating LLMs and retrieval systems, evaluating model behavior, and deploying reliable automation for real operational use cases.

Experienced in forward-deployed AI engineering, including stakeholder discovery, workflow mapping, rapid prototyping, customer-facing implementation, technical documentation, evaluation design, and translating ambiguous business problems into working AI systems. Focused on building practical AI tools that improve knowledge access, reduce manual work, automate repetitive workflows, and make complex business processes easier to complete.

Core Skills

AI Engineering	LLM Applications, AI Agents, RAG Systems, Retrieval-Augmented Generation, Prompt Engineering, Structured Outputs, Function Calling, Tool-Using Agents, Human-in-the-Loop AI, AI Workflow Automation, Document Intelligence
RAG & Knowledge Systems	Embeddings, Vector Search, FAISS, Chroma, Pinecone, Weaviate, Re-ranking, Knowledge Bases, Context Engineering, Retrieval Evaluation, Grounded Answer Generation
AI Reliability & Evaluation	LLM Evaluation, Prompt Evaluation, Regression Testing, Failure-Mode Analysis, Hallucination Checks, Retrieval Quality Measurement, Semantic Similarity, Token-Level Accuracy, Model Comparison, Feedback Loops
Backend & Implementation	Python, FastAPI, REST APIs, WebSockets, SQL, PostgreSQL, SQL Server, SQLite, Authentication, Logging, Docker, CI/CD, API Integration, Data Pipelines
Forward-Deployed Engineering	Stakeholder Discovery, Workflow Mapping, Requirement Translation, Technical Scoping, MVP Delivery, Customer Implementation, Documentation, Training, AI Adoption, Process Standardization
AI Development Tools	LangChain, LlamaIndex, Hugging Face Transformers, PyTorch, Cursor, Claude Code, LangSmith, Weights & Biases, OpenAI, Claude, Llama Models

Professional Experience

Graduate Research Assistant / AI-ML Research Engineer – LLM Systems Aug 2024 – Present
Wright State University *Dayton, OH | Part-time | On-site*

- Built reusable LLM evaluation workflows for reasoning quality, retrieval performance, ranking behavior, failure-mode analysis, and generalization across unseen input patterns.
- Developed structured evaluation pipelines to measure token-level accuracy, semantic usefulness, ranking quality, edit similarity, and model robustness.
- Improved experiment repeatability through preprocessing, ablation studies, model comparison, prompt/context refinement, and standardized reporting workflows.
- Worked on AI/ML research involving code intelligence, reverse engineering, function-name inference, transformer-based models, and structured LLM output evaluation.
- Translated research findings into implementation improvements, including better edge-case handling, stronger context construction, refined prompts, and clearer evaluation criteria.
- Designed repeatable analysis workflows for evaluating model behavior, identifying failure patterns, and improving AI system reliability.

- Supported research-to-production thinking by converting experimental model results into measurable quality benchmarks, diagnostics, and implementation recommendations.

Key Technologies: Python, Transformers, LLM Evaluation, Prompt Evaluation, Machine Learning, Code Intelligence, Model Analysis, AI Reliability

Senior AI Engineer – LLM Applications & Agent Workflows
PROCIT BV

Feb 2022 – Aug 2024
Netherlands | Full-time | Remote

- Built production-oriented LLM applications, RAG assistants, AI workflow automations, and API-connected services for support, reporting, CRM workflows, customer communication, document processing, and business operations.
- Developed Python/FastAPI backends with LLM integrations, retrieval pipelines, business rules, authentication, logging, auditability, data pipelines, and product logic.
- Designed and implemented RAG-based knowledge assistants for troubleshooting, internal knowledge access, support automation, document search, and operational workflows.
- Converted scattered documentation, troubleshooting notes, and process knowledge into searchable AI workflows using embeddings, vector search, re-ranking, structured prompts, and feedback loops.
- Built evaluation and regression workflows to measure response quality, detect weak retrieval, identify prompt failures, and guide production improvements.
- Improved AI reliability by adding human feedback loops, approval checkpoints, execution traces, structured output validation, and failure-handling patterns.
- Reduced unnecessary LLM usage through retrieval filtering, caching, re-ranking, better prompt design, and context optimization.
- Automated document-processing workflows involving extraction, validation, routing, communication, and business-process handoffs.
- Improved prototype turnaround by rapidly scoping workflows, validating feasibility, and hardening successful AI experiments into reusable systems.
- Worked directly with stakeholders to convert ambiguous business needs into AI prototypes, technical requirements, deployment plans, and production-ready workflow automation.

Key Technologies: Python, FastAPI, LLMs, RAG, LangChain, Vector Search, Embeddings, REST APIs, SQL, Prompt Engineering, Workflow Automation, AI Evaluation

Software Engineer / Automation Developer
Independent, Startup, and Enterprise AI/Software Projects

Jan 2018 – Jan 2022
Remote / Hybrid | Contract

- Delivered 10+ backend, automation, dashboard, and data-processing projects across startup, enterprise, and independent environments.
- Built automation workflows using Python, SQL databases, REST APIs, web applications, data pipelines, and integration logic to reduce manual business processes.
- Developed backend services, operational dashboards, reporting workflows, validation systems, and user-facing automation tools.
- Connected fragmented systems through SQL databases, APIs, Python automation, scripts, and web applications to improve data flow and operational efficiency.
- Improved workflow reliability through debugging, data validation, stakeholder communication, process mapping, and technical documentation.
- Built the software engineering foundation for later AI engineering work involving LLM systems, RAG, agents, workflow automation, and production AI implementation.
- Supported business users by translating workflow problems into technical solutions, automation scripts, and practical backend tools.

Key Technologies: Python, SQL, REST APIs, Backend Development, Automation, Data Pipelines, Dashboards, System Integration, Technical Documentation

Selected AI Projects

Smart365Guide – AI-Guided Workflow Assistant

LLM Guidance | Workflow Automation |

Enterprise Software Support

- Designed an AI-powered guidance layer that helps users understand screens, complete complex workflows, and receive contextual support inside web and enterprise applications.
- Built product concepts around Explain Mode, Guide Mode, and action-oriented AI assistance for onboarding, workflow completion, training, and support reduction.
- Designed workflow-aware assistance patterns using app context, structured process knowledge, user intent, and human-in-the-loop controls.
- Focused on making complex SaaS and ERP workflows easier to understand, navigate, and complete through AI-guided interaction.

Technologies: LLMs, RAG, Workflow Automation, JavaScript SDK Concepts, Product Design, AI Guidance Systems

MomoBot – Agentic Workflow Automation Platform

Python | FastAPI | WebSockets | Agents |

RBAC

- Built a working agentic workflow automation prototype with task orchestration, tool execution, scheduling, approval gates, execution traces, and audit logs.
- Designed human-in-the-loop execution patterns for safe automation, monitoring, task dispatch, and business workflow control.
- Improved reliability for long-horizon automation through REST/WebSocket execution loops, policy gates, role-based controls, and reusable workflow patterns.
- Focused on secure AI automation with approval-based execution instead of uncontrolled autonomous actions.

Technologies: Python, FastAPI, WebSockets, Agents, RBAC, Workflow Automation, Audit Logs, Human-in-the-Loop AI

Domain-Specific RAG & Document Intelligence Assistants

Embeddings | Vector Search |

Re-ranking | Document AI

- Built retrieval-grounded assistants that turn unstructured documents, troubleshooting notes, and operational knowledge into searchable AI systems.
- Implemented RAG patterns using embeddings, vector search, re-ranking, structured extraction, and feedback loops.
- Improved answer consistency by detecting weak retrieval, refining context construction, and reducing unreliable responses.
- Designed reusable RAG workflows for internal support, onboarding, document-heavy workflows, and operational knowledge access.

Technologies: RAG, Embeddings, FAISS, Chroma, Pinecone, Vector Search, Re-ranking, LLMs, Document Intelligence

ReFuN – Reasoning-Guided LLM Function Name Inference

Transformers | Code Intelligence |

LLM Evaluation

- Built reasoning-guided LLM workflows for function-name inference from stripped binaries using assembly, decompiled code, and structured program representations.
- Evaluated AI output quality using exact match, token F1, edit similarity, semantic quality, and failure-mode analysis.
- Applied transformer-based modeling and structured evaluation methods to code intelligence and reverse engineering tasks.
- Converted experimental model behavior into evaluation criteria, failure diagnostics, and implementation improvements.

Technologies: Python, Transformers, Code Intelligence, LLM Evaluation, Reverse Engineering, Machine Learning

Technical Stack

Languages	Python, SQL, TypeScript/JavaScript, C#, Bash
Backend	FastAPI, REST APIs, WebSockets, Node.js/Express, Authentication, Logging, Docker, CI/CD
LLMs	GPT-4o, GPT-4 Turbo, Claude, Llama, Hugging Face Transformers
RAG / Agents	LangChain, LlamaIndex, Embeddings, Vector Search, FAISS, Chroma, Pinecone, Weaviate, Re-ranking, Function Calling
AI Evaluation	Prompt Evaluation, Regression Testing, Failure Analysis, Retrieval Evaluation, Quality Measurement, Model Comparison
Data	SQL Server, PostgreSQL, SQLite, Data Pipelines, Structured Extraction, Reporting Workflows
Product / Delivery	MVP Planning, Workflow Mapping, Stakeholder Interviews, Documentation, AI Adoption, User Feedback Loops

Education

M.S. in Computer Science – AI / AI Agents / Cybersecurity Focus <i>Wright State University</i>	<i>Dayton, OH</i>
B.E. in Computer Engineering <i>Pokhara University</i>	<i>Nepal</i>

Awards & Recognition

- **Shortest Customer Onboarding Record – PROCIT BV:** Supported one of the fastest customer onboarding cycles by helping move a customer from setup to working implementation within 3 weeks.
- **Best Employee – PROCIT BV:** Recognized for strong ownership, technical delivery, customer-focused problem solving, and contribution to AI/software implementation projects.
- **Rapid AI Prototype Delivery Recognition:** Delivered working AI workflow prototypes within short implementation cycles by quickly translating business requirements into usable LLM/RAG systems.
- **Customer Workflow Automation Impact:** Recognized for improving operational workflows through automation, internal dashboards, API integrations, document-processing pipelines, and AI-assisted business processes.
- **AI Implementation Excellence:** Contributed to practical deployment of AI systems involving LLM integrations, RAG workflows, human-in-the-loop controls, structured outputs, and workflow automation.
- **First Prize – ICT Fest:** Won first prize for an AI-Based Nepali Sign Language Recognition project.
- **Research Contributions:** Research papers under review: ReFuN; Learning from Errors: Deviation Map-Augmented Feature Representation Learning for Multi-Modal Aerial Imagery.